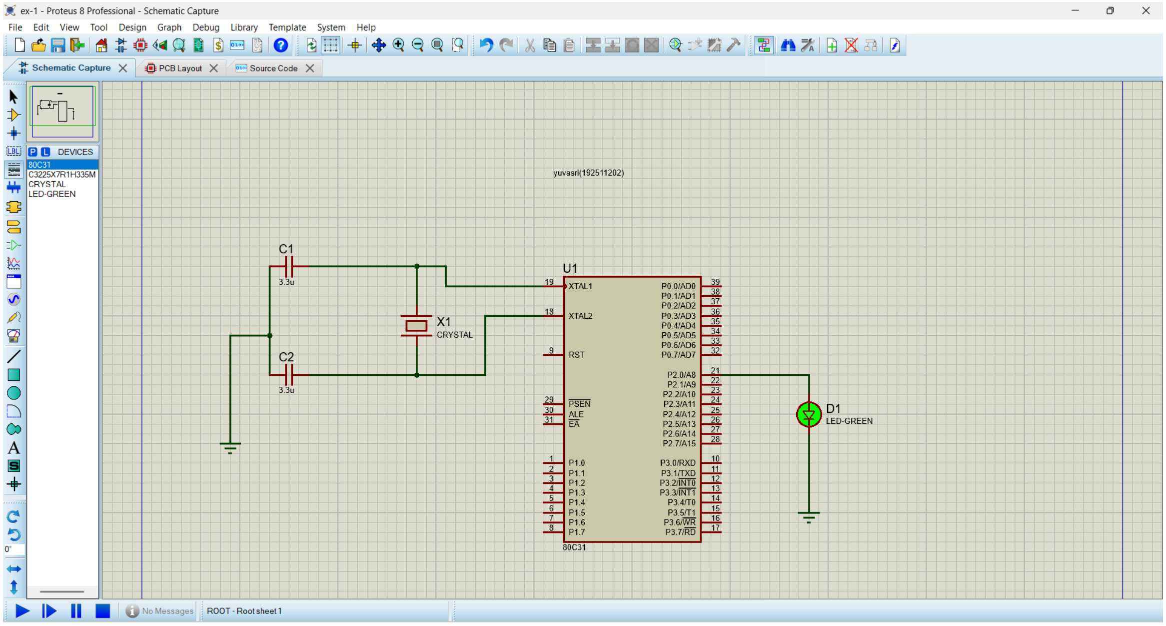
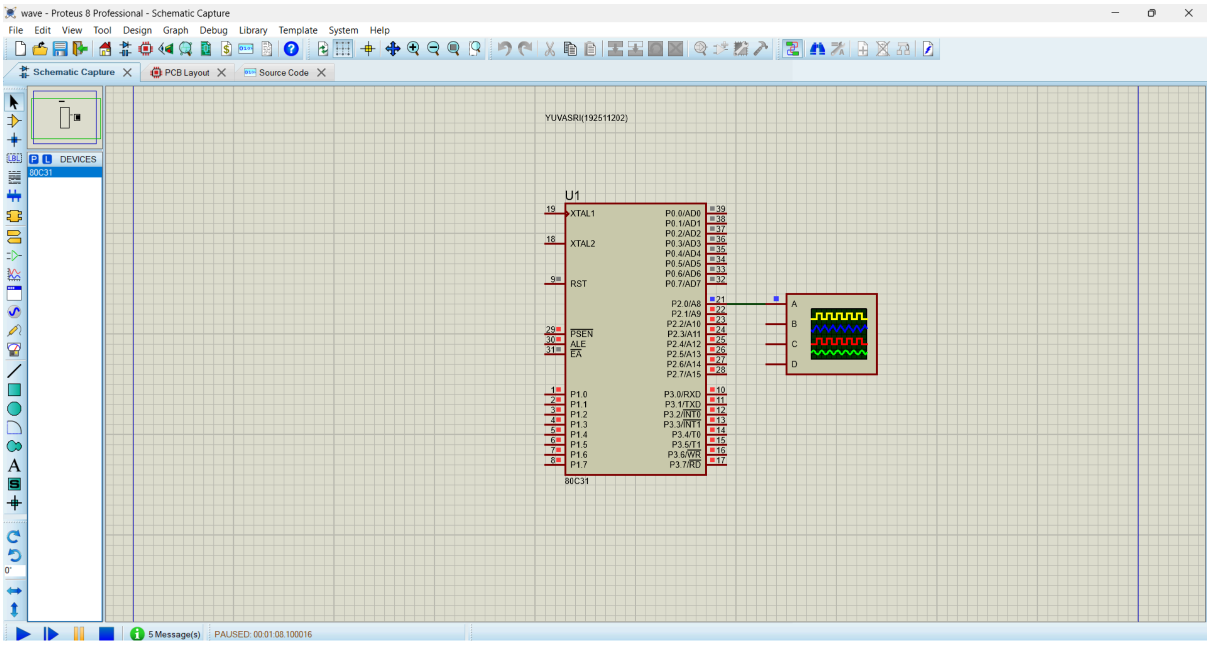
ECA14 – EMBEDDED SYSTEMS

List of Experiments

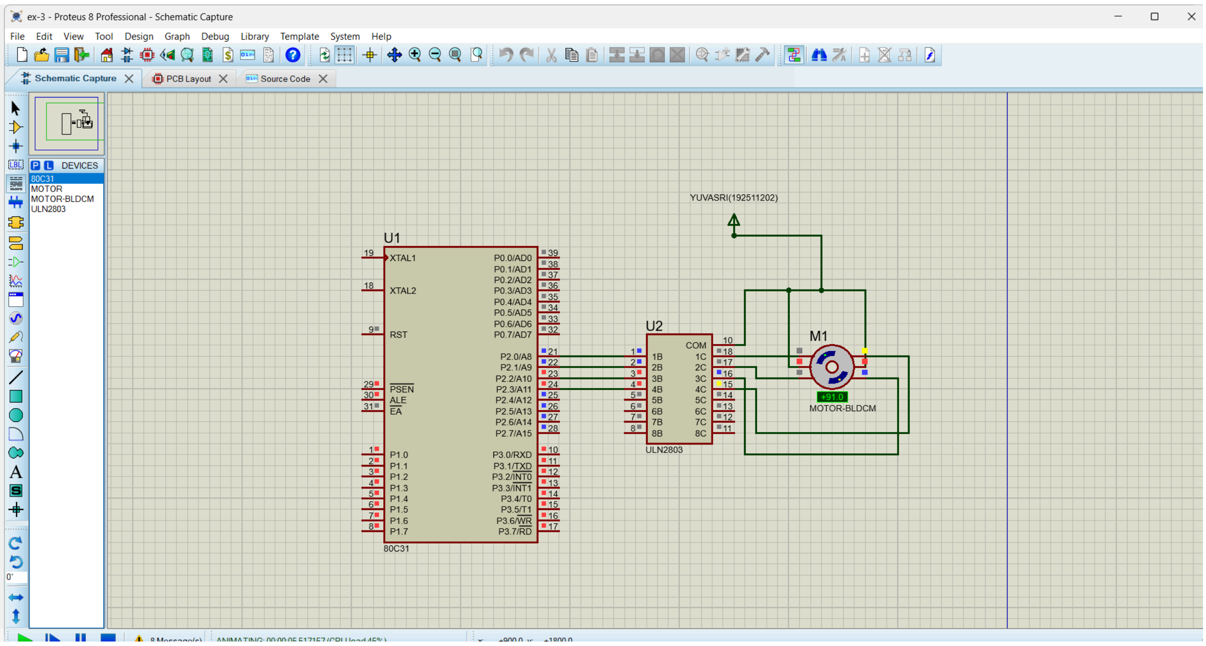
EX-1. 1 Flashing of Led using AT89C51 Microcontroller using Proteus



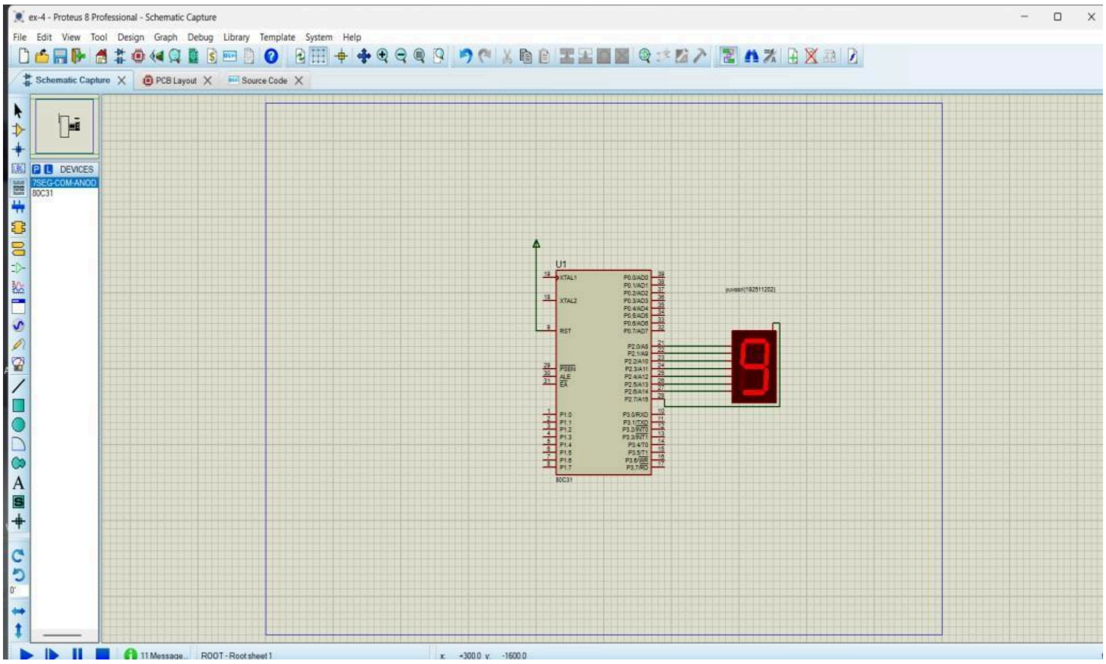
EX-2 Generation of Square Wave using Proteus



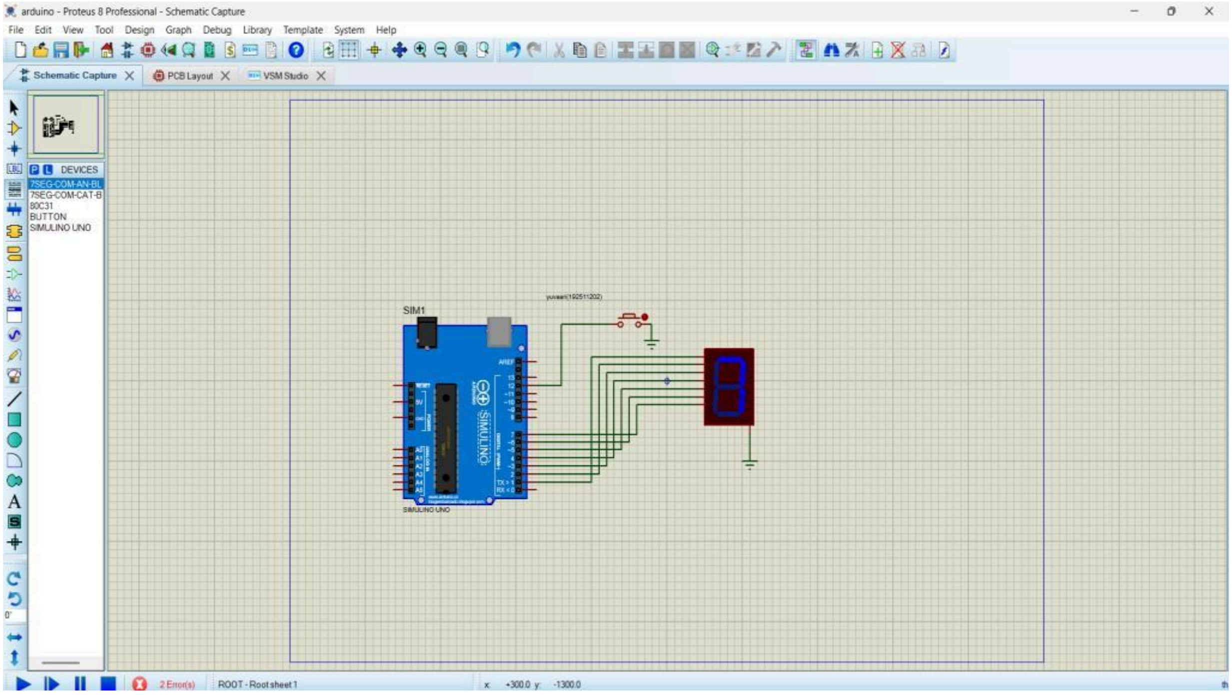
EXP 3: STEPPER MOTOR USING AT89C51 USING PROTEUS



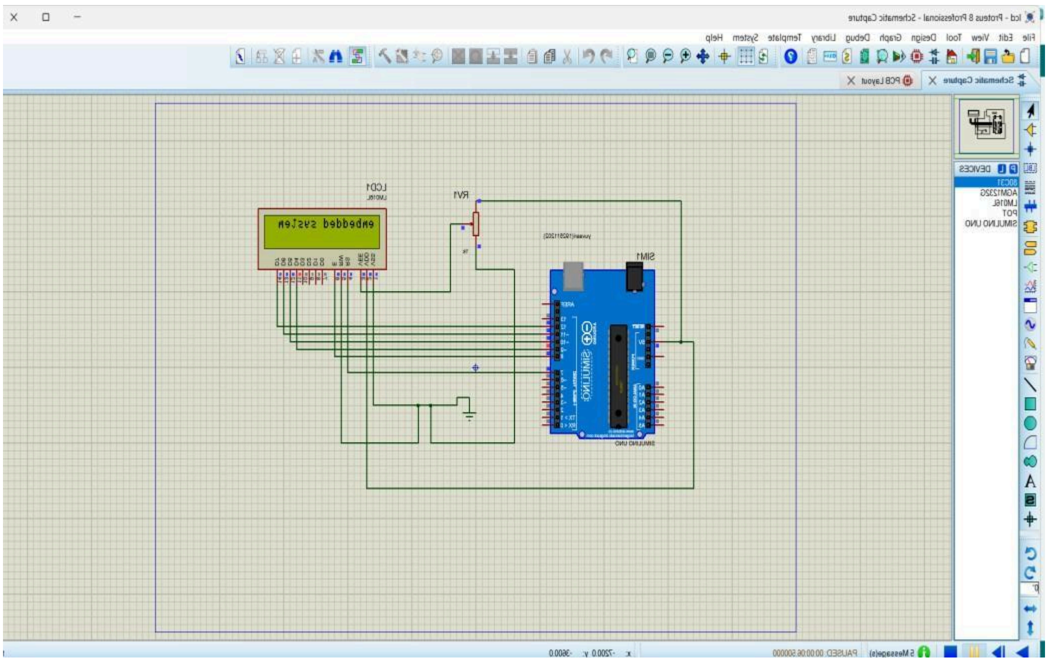
EXP 4: 7 SEGMENT DISPLAY USING AT89C51 USING PROTEUS



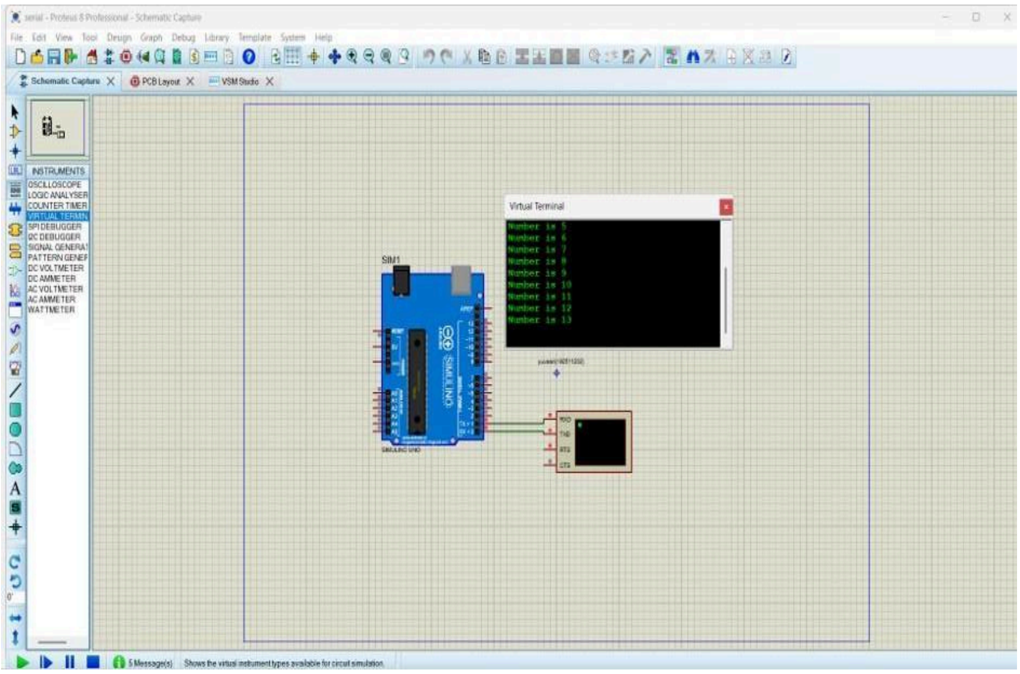
EXP 5: ARDUINO BASED MANUAL ELECTRONIC COUNTER USING PROTEUS



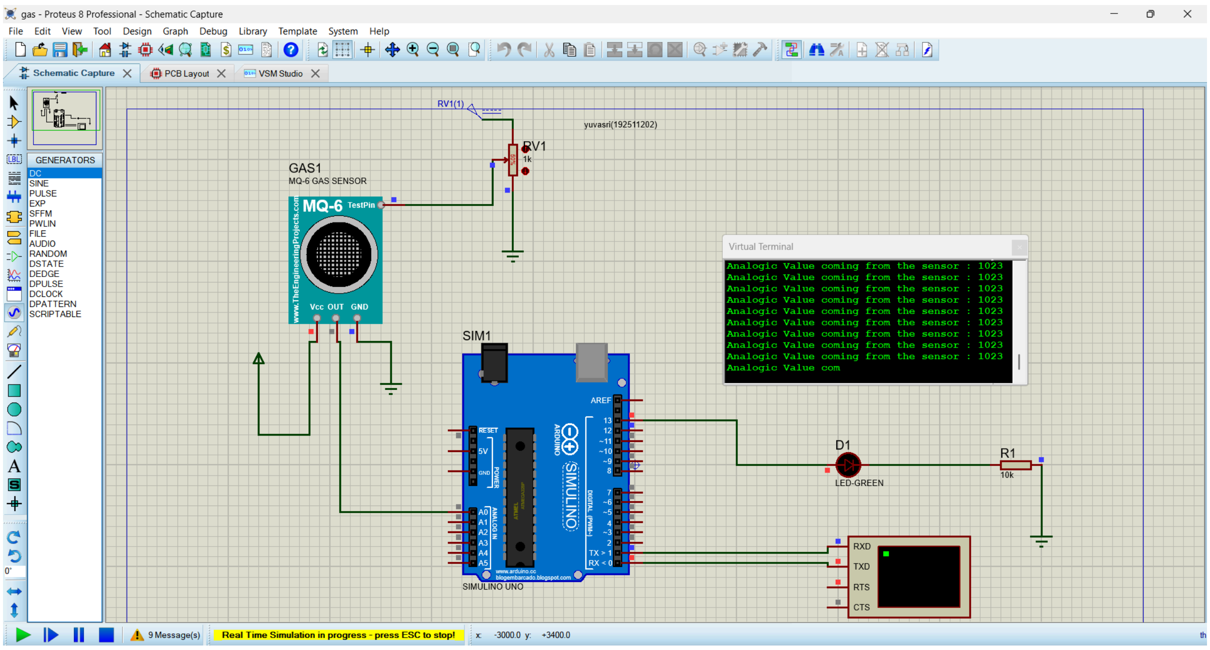
EXP 6 : ARDUINO TO 16x2 LCD DISPLAY USING PROTEUS



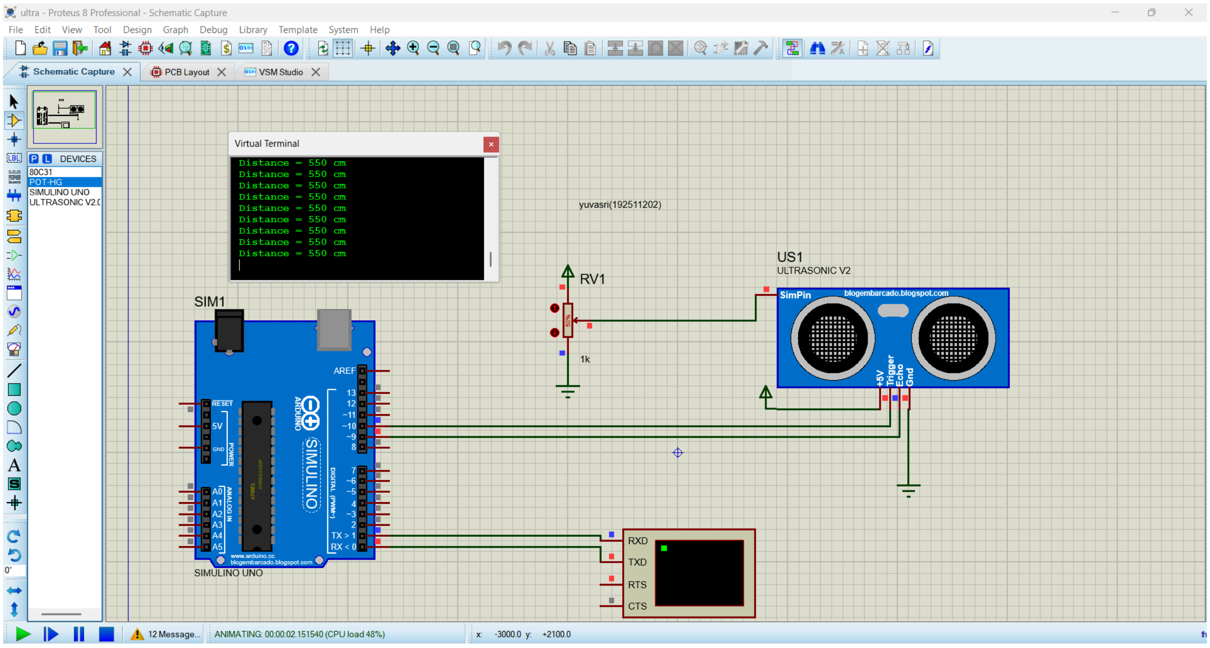
EXP 7: SERIAL COMMUNICATION USING ARDUINO WITH PROTEUS



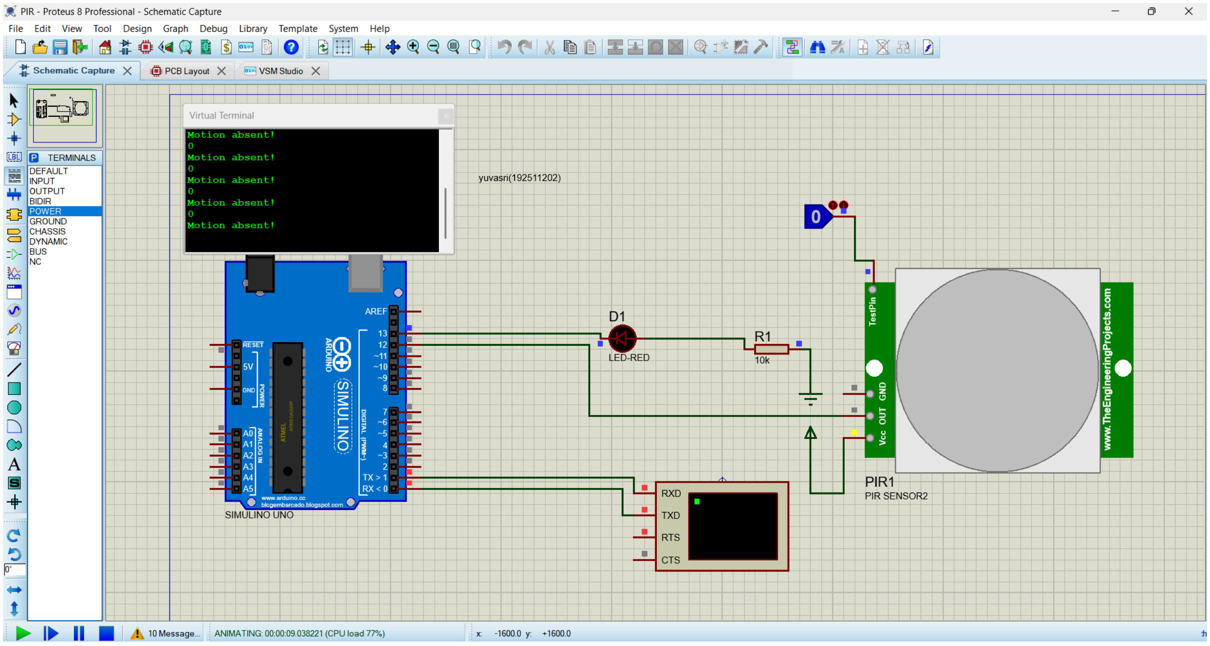
EXP 8 : GAS SENSOR MQ-2 IN PROTEUS USING ARDUINO



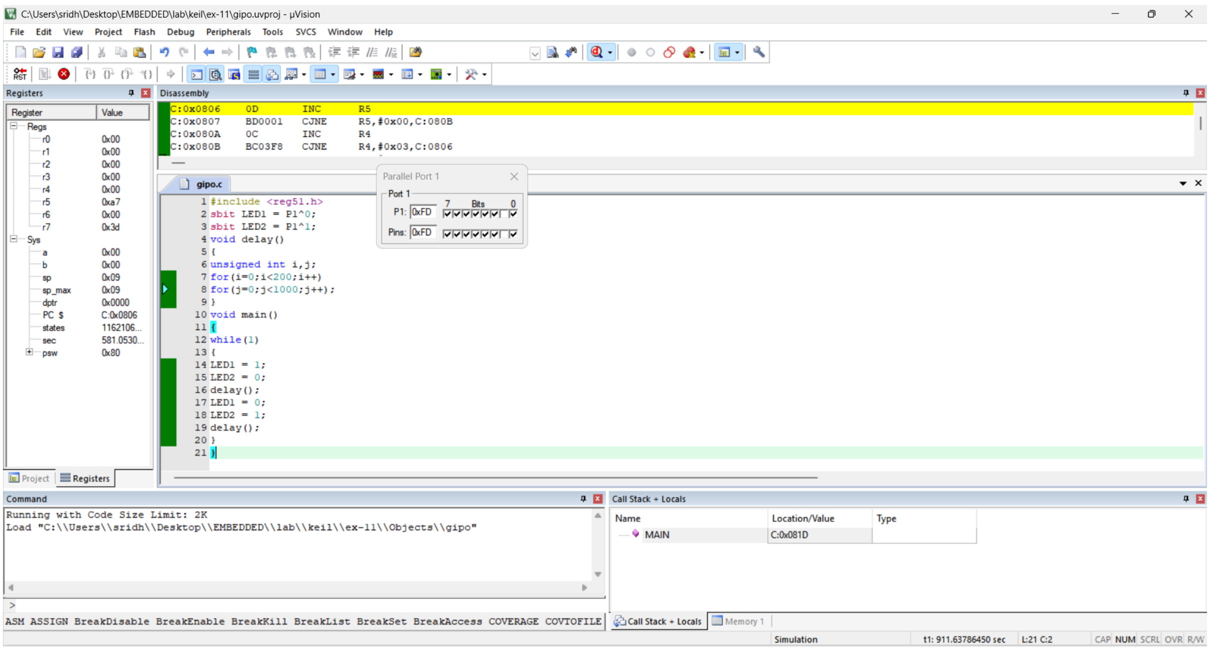
EXP 9 : ULTRASONIC SENSOR IN PROTEUS USING ARDUINO



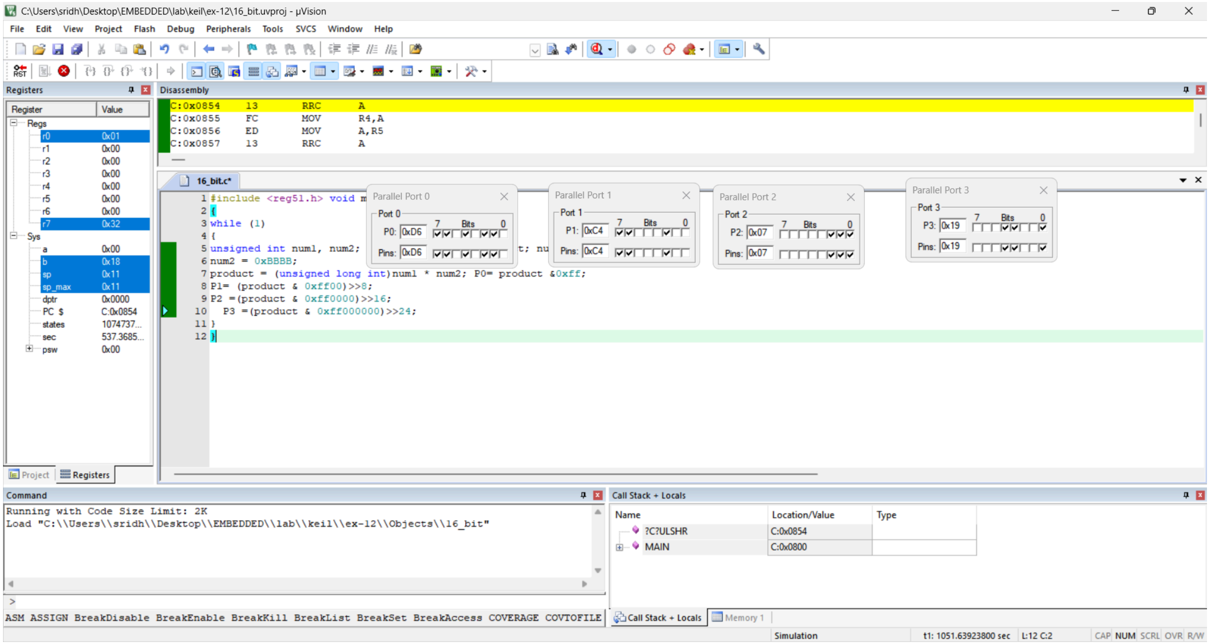
EXP 10 : PIR SENSOR IN PROTEUS USING ARDUINO



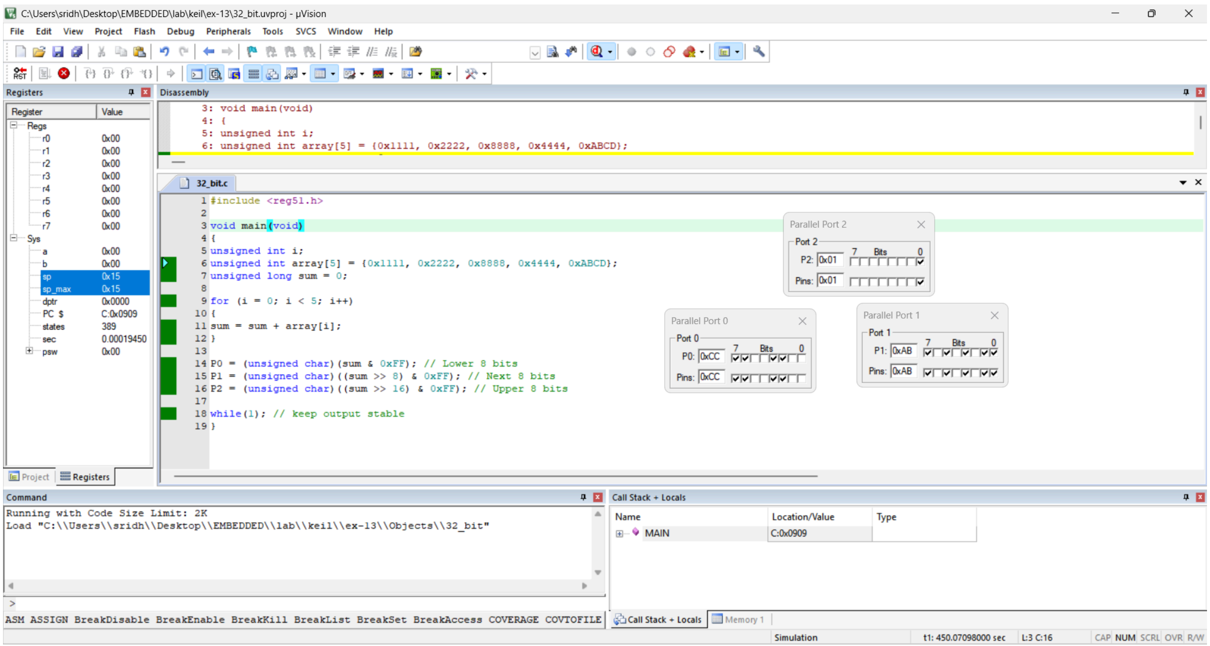
EXP 11 : STUDY OF GPIO & BIT ADDRESSING IN AT89C51



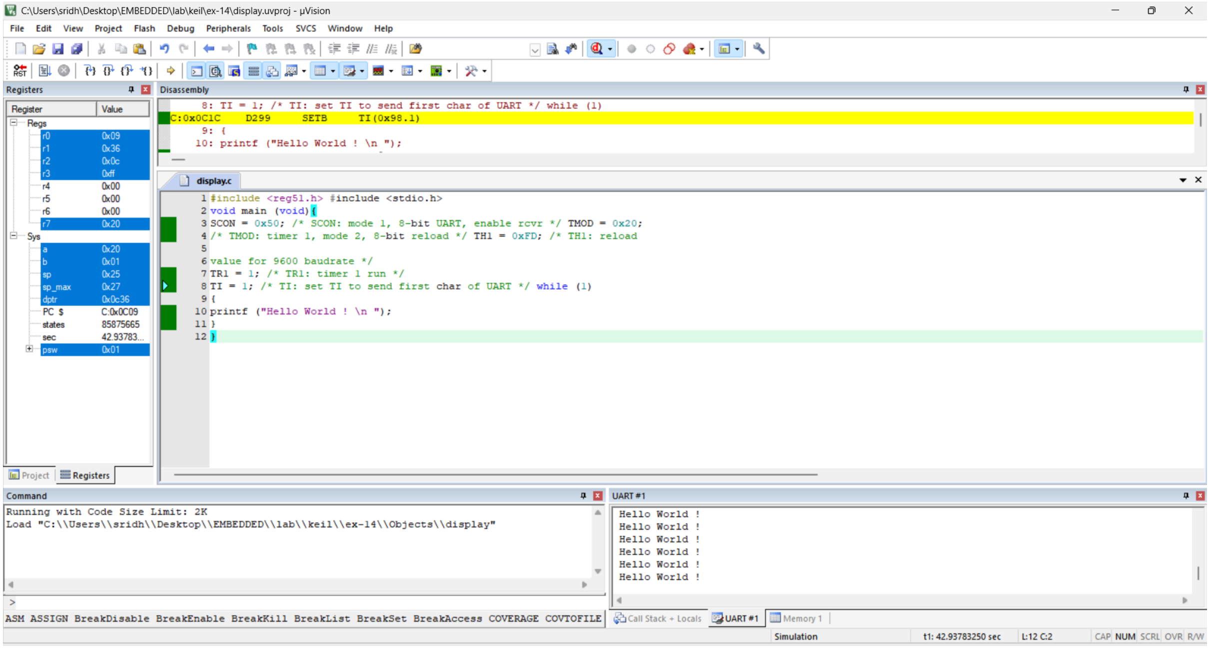
EXP 12: EMBEDDED C PROGRAM TO MULTIPLY TWO 16-BIT BINARY NUMBERS



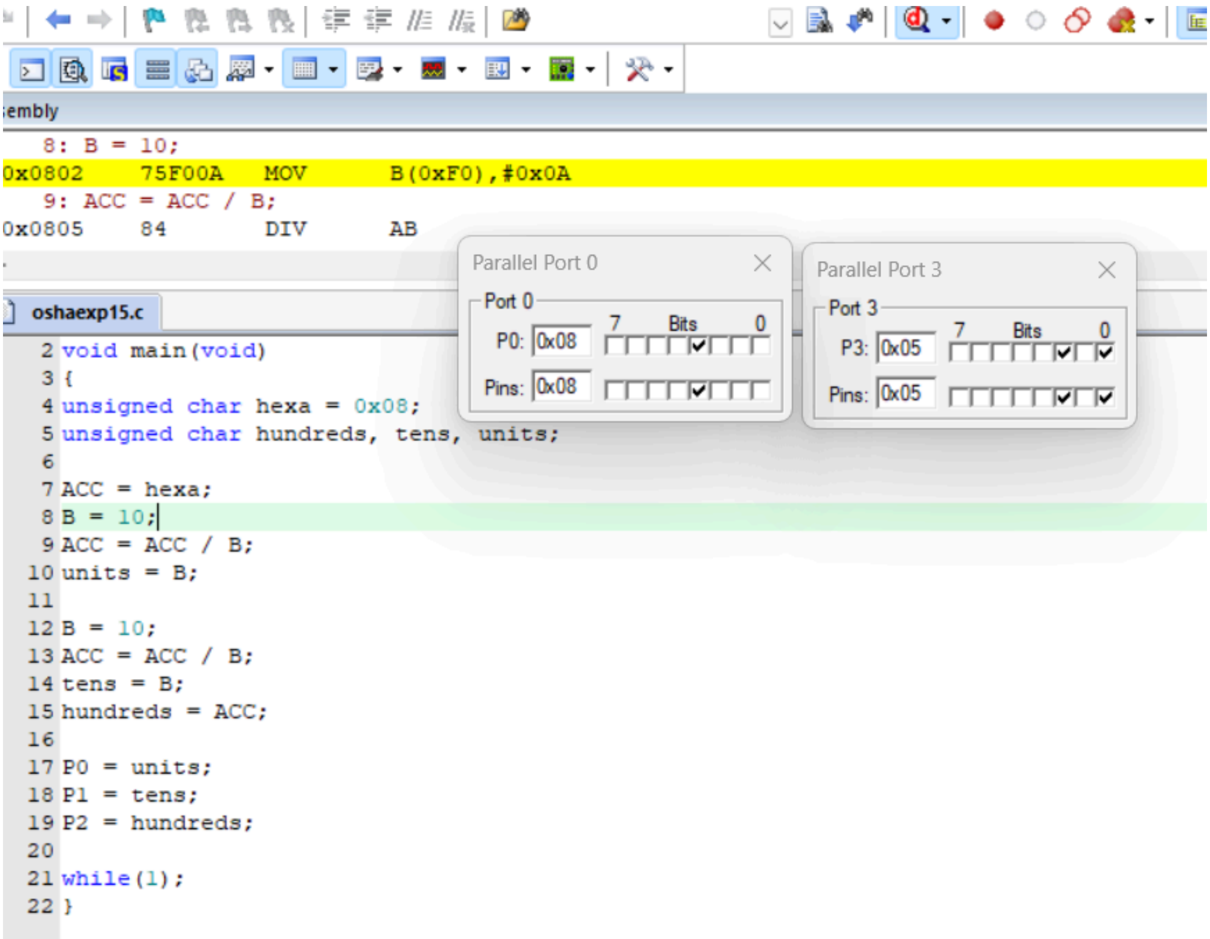
EXP 13: EMBEDDED C PROGRAM TO ADD AN ARRAY OF 16-BIT NUMBERS ANDSTORE THE 32-BIT RESULT IN INTERNAL RAM



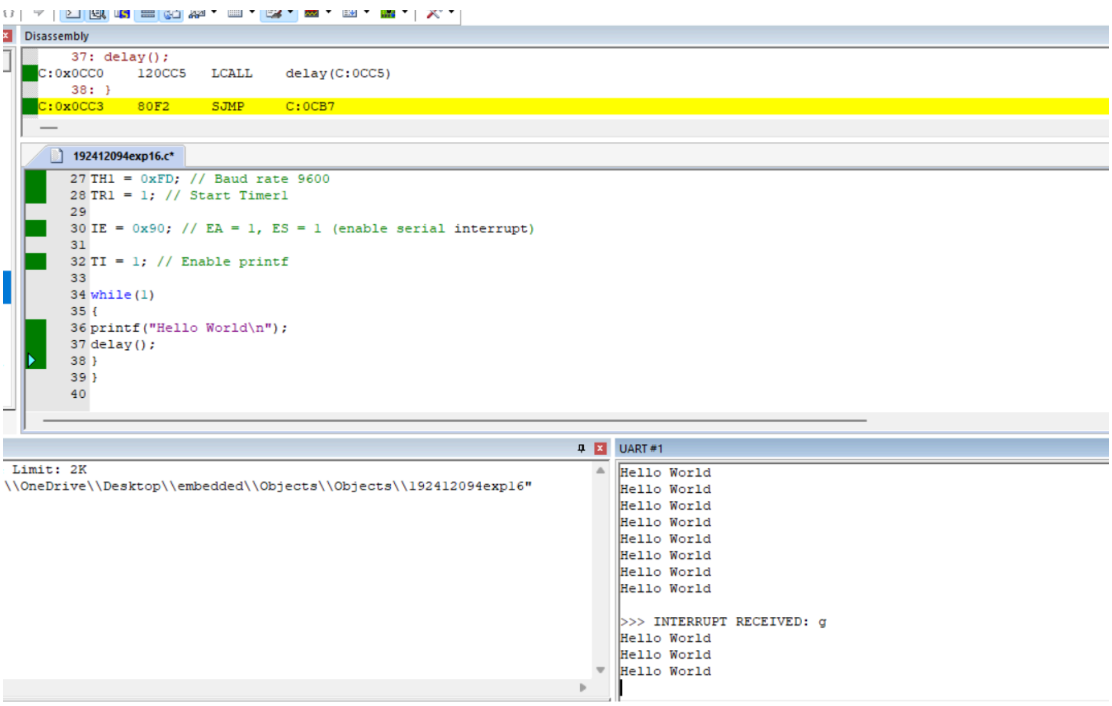
EXP 14: EMBEDDED C PROGRAM TO DISPLAY “HELLO WORLD” MESSAGE



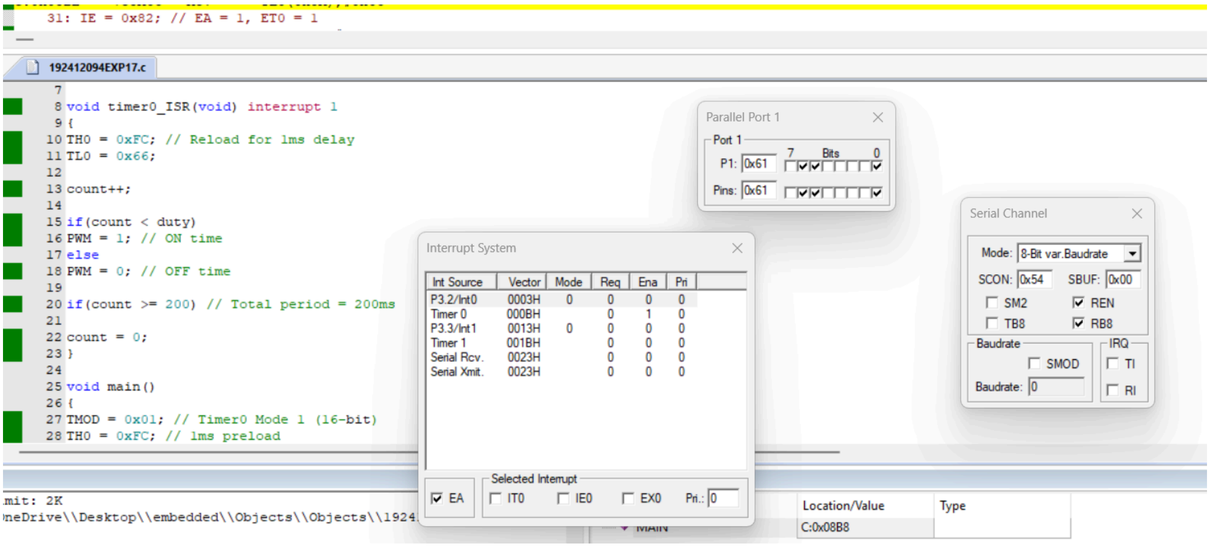
EXP 15: EMBEDDED C PROGRAM TO CONVERT THE HEXADECIMAL TO DECIMAL AND DISPLAY THE DIGITS ON PORTS P0, P1 AND P2



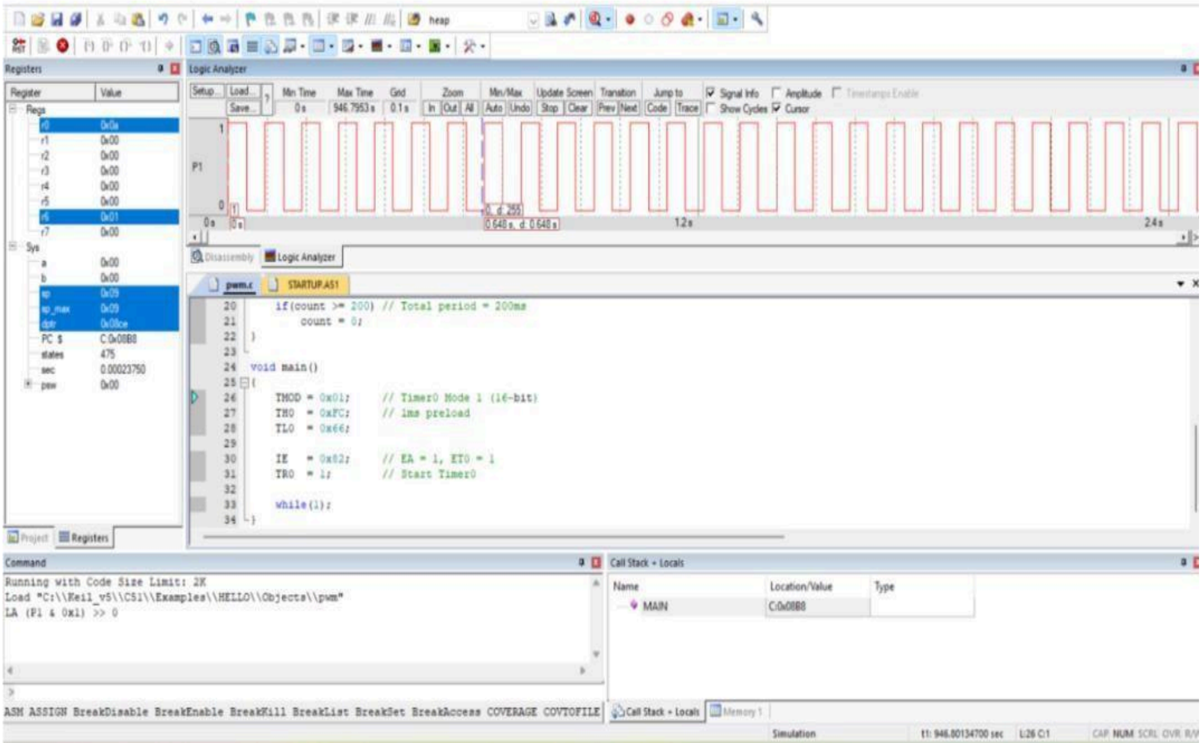
EXP16 : SERIAL INTERRUPT USING UART (KEYBOARD INPUT) IN KEIL

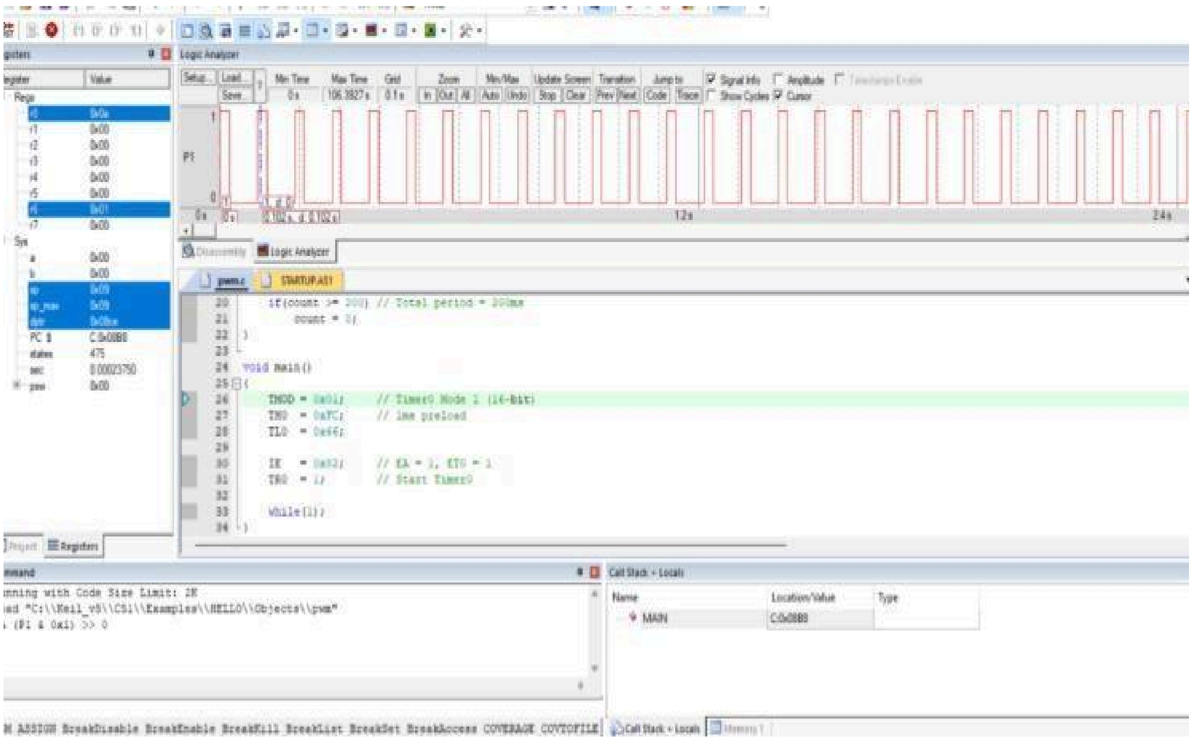


EXP 17: UART INTERFACING USING INTERRUPT IN KEIL

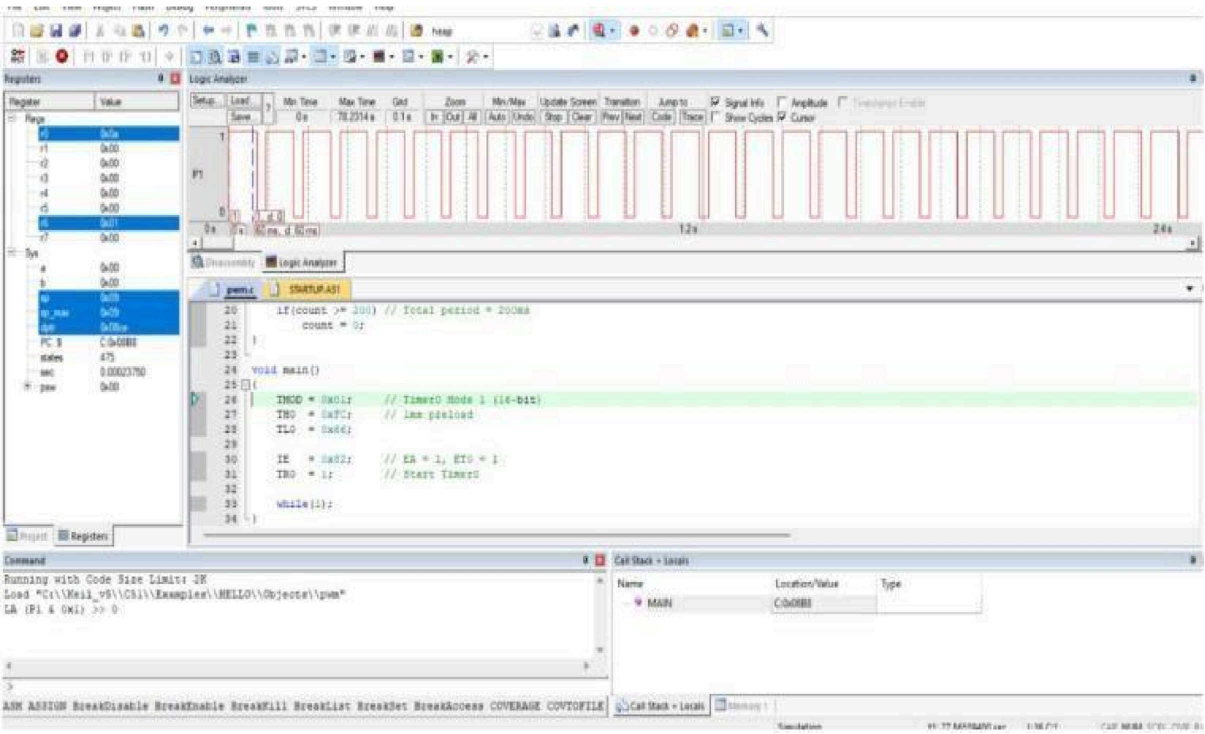


EXP 18: Generation of pwm using timer interrupt





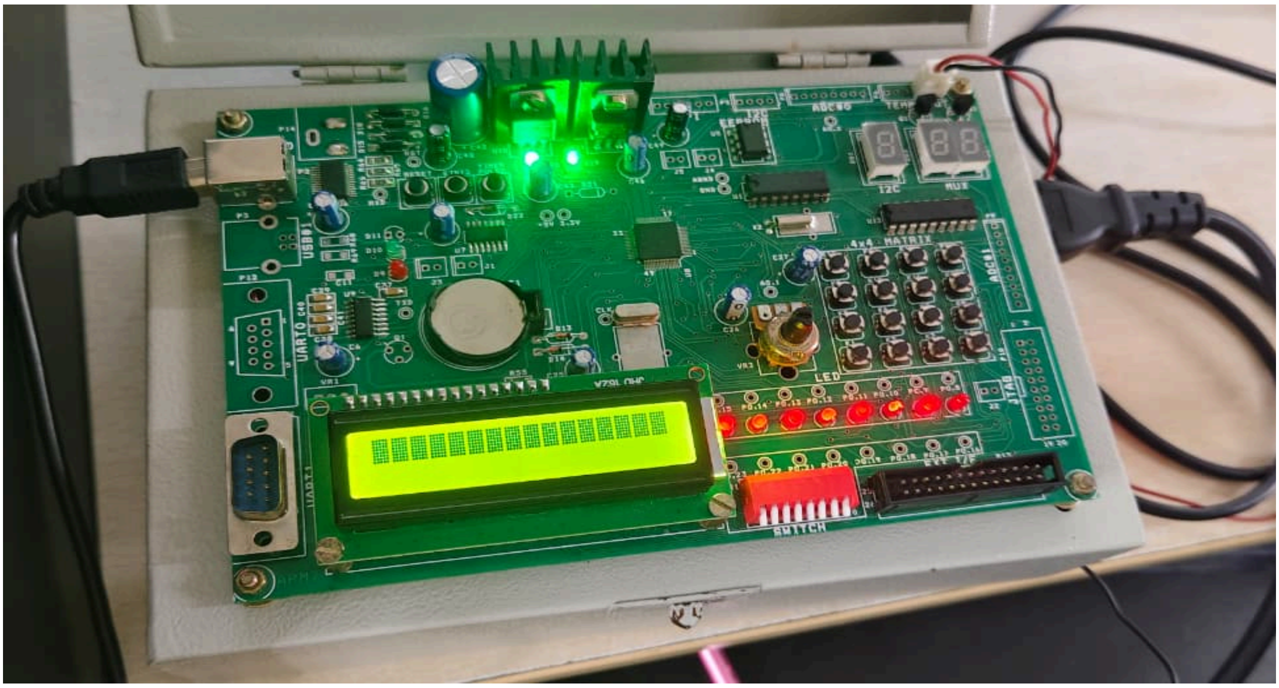
Logic Analyzer
P1
if(count >= 200) // Total period = 200ms
count = 0;
void main()
TMOD = 0x01; // Timer0 Mode 1 (16-bit)
TH0 = 0xFC; // 1ms preload
TL0 = 0x66;
IE = 0x82; // EA = 1, ET0 = 1
TR0 = 1; // Start Timer0
while(1);
Call Stack + Locals
MAIN



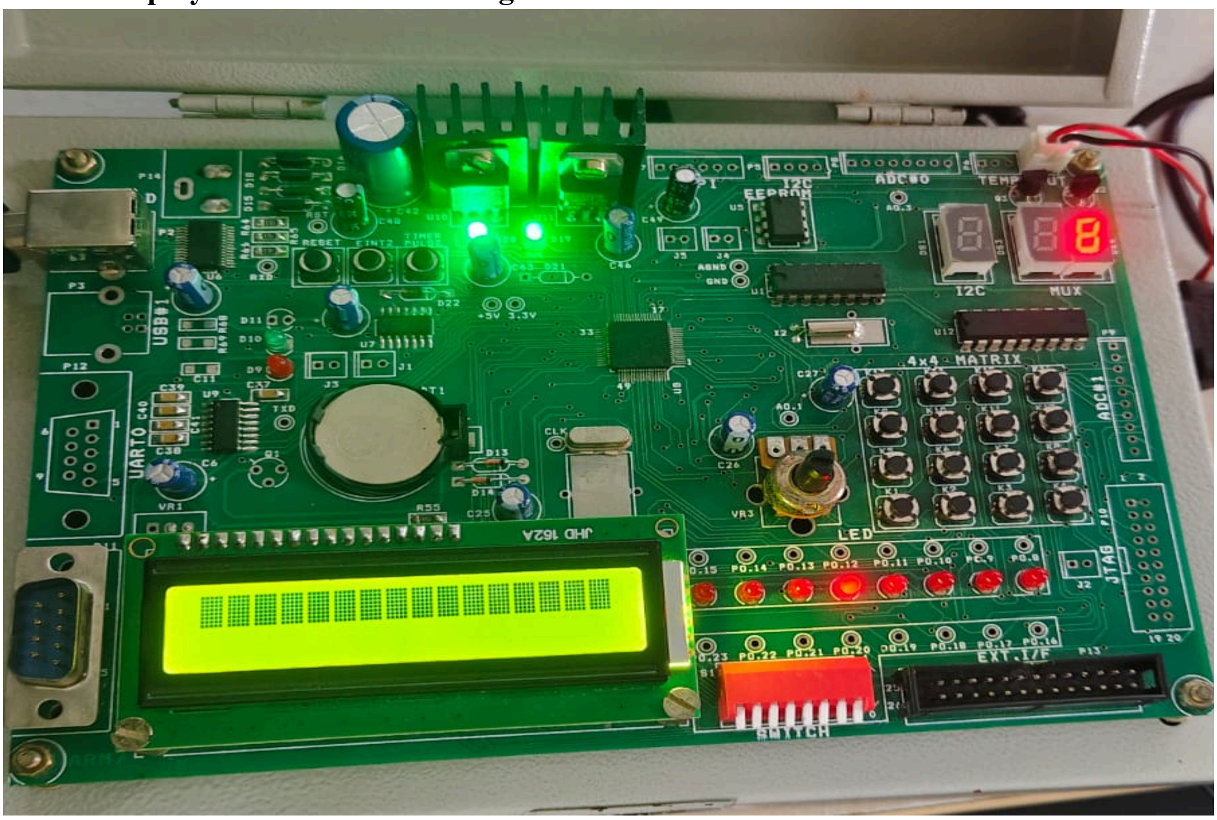
Logic Analyzer
P1
if(count >= 200) // Total period = 200ms
count = 0;
void main()
TMOD = 0x01; // Timer0 Mode 1 (16-bit)
TH0 = 0xFC; // 1ms preload
TL0 = 0x66;
IE = 0x82; // EA = 1, ET0 = 1
TR0 = 1; // Start Timer0
while(1);
Running with Code Size Limit: 2K
LA (P1 & 0x1) >> 0
Call Stack + Locals
MAIN

EXP 19: Blinking LEDs using software delay routine in LPC2148 kit

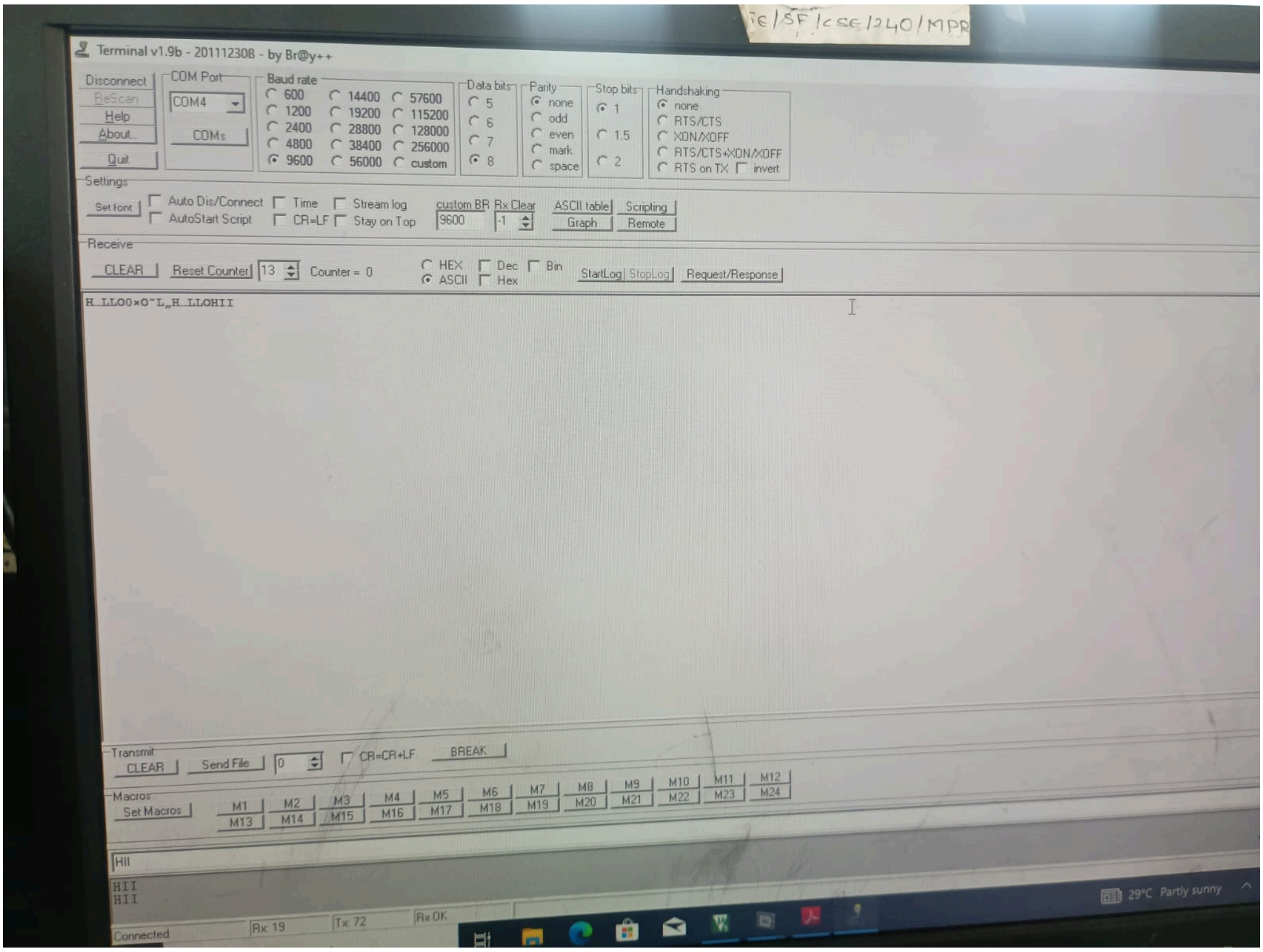
EX 20:Program to read the switch and display in the LEDs using LPC2148 kit



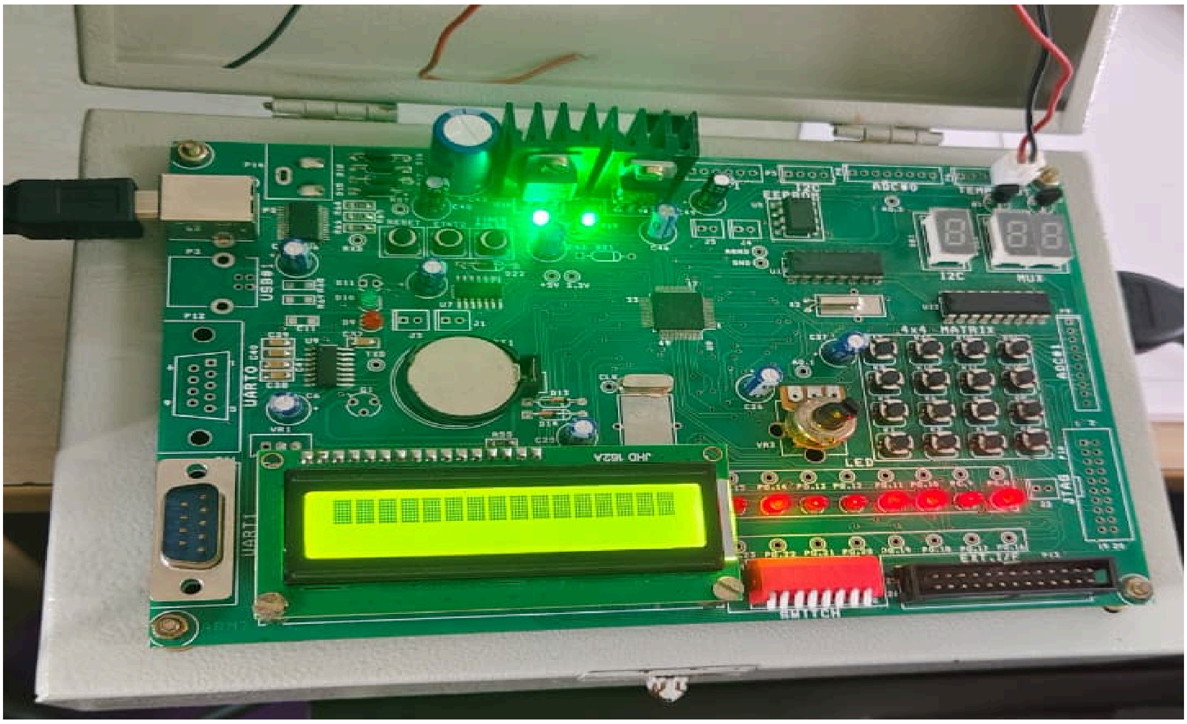
EX 21: Display a number in seven segment LED in LPC2148 kit



EX 22:Serial transmission and reception using on-chip UART in LPC2148 kit



EX 23:Access an internal ADC and display the binary output in LEDs in LPC2148 kit



Exp 24: Modeling Clock-Driven Concurrent Processes Using SC_THREAD and wait() in SystemC

c

```
saisri@DESKTOP-P73175QB:~/systemc_test/build$ ./test_systemc

        SystemC 2.3.4-Accellera --- May 20 2025 11:42:17
        Copyright (c) 1996-2022 by all Contributors,
        ALL RIGHTS RESERVED
Time: 0 s - Clock Triggered
Time: 1 ns - Clock Triggered
Time: 2 ns - Clock Triggered
Time: 3 ns - Clock Triggered
Time: 4 ns - Clock Triggered
saisri@DESKTOP-P73175QB:~/systemc_test/build$
```

EXP 25: Producer–Consumer Problem Using SystemC

```
saisri@DESKTOP-P73175QB:~/systemc_test/build$ cd ~/systemc_test
saisri@DESKTOP-P73175QB:~/systemc_test$ cd build
saisri@DESKTOP-P73175QB:~/systemc_test/build$ cmake ..
-- Configuring done (0.0s)
-- Generating done (0.0s)
-- Build files have been written to: /home/saisri/systemc_test/build
saisri@DESKTOP-P73175QB:~/systemc_test/build$ make
[ 33%] Built target and_gate
[ 66%] Built target producer_consumer
[100%] Built target test_systemc
saisri@DESKTOP-P73175QB:~/systemc_test/build$ ./producer_consumer

        SystemC 2.3.4-Accellera --- Aug 21 2026 05:34:04
        Copyright (c) 1996-2022 by all Contributors,
        ALL RIGHTS RESERVED
Produced: 1
Consumed: 1
Produced: 2
Consumed: 2
Produced: 3
Consumed: 3
Produced: 4
Consumed: 4
Produced: 5
Consumed: 5
saisri@DESKTOP-P73175QB:~/systemc_test/build$
```

EXP 26: Transaction Level Modeling (TLM) using SystemC

```
saisri@DESKTOP-P73175QB:~/systemc_test/build$ cd ..
saisri@DESKTOP-P73175QB:~/systemc_test$ ls
CMakeLists.txt  and_gate.cpp  build  producer_consumer.cpp  test_systemc.cpp  test_systemcc.cpp
saisri@DESKTOP-P73175QB:~/systemc_test$ nano tlm.cpp
saisri@DESKTOP-P73175QB:~/systemc_test$ nano CMakeLists.txt
saisri@DESKTOP-P73175QB:~/systemc_test$ cd build
saisri@DESKTOP-P73175QB:~/systemc_test/build$ cmake ..
-- Configuring done (0.0s)
-- Generating done (0.0s)
-- Build files have been written to: /home/saisri/systemc_test/build
saisri@DESKTOP-P73175QB:~/systemc_test/build$ make
[ 25%] Built target and_gate
[ 50%] Built target producer_consumer
[ 75%] Built target test_systemc
[ 87%] Building CXX object CMakeFiles/tlm.dir/tlm.cpp.o
[100%] Linking CXX executable tlm
[100%] Built target tlm
saisri@DESKTOP-P73175QB:~/systemc_test/build$ ./tlm
    SystemC 2.3.4-Accellera --- Aug 21 2026 05:34:04
    Copyright (c) 1996-2022 by all Contributors,
    ALL RIGHTS RESERVED
Initiator sending: 555
Target received: 555
saisri@DESKTOP-P73175QB:~/systemc_test/build$
```

EXP 27: Binary Semaphore in SystemC

```
saisri@DESKTOP-P73175QB:~/systemc_test$ nano CMakeLists.txt
saisri@DESKTOP-P73175QB:~/systemc_test$ ls /opt/systemc
include  lib  share
saisri@DESKTOP-P73175QB:~/systemc_test$ ls /opt/systemc/lib
cmake  libsystemc.so  libsystemc.so.2.3  libsystemc.so.2.3.4
saisri@DESKTOP-P73175QB:~/systemc_test$ nano binary_semaphore.cpp
saisri@DESKTOP-P73175QB:~/systemc_test$ nano CMakeLists.txt
saisri@DESKTOP-P73175QB:~/systemc_test$ nano CMakeLists.txt
saisri@DESKTOP-P73175QB:~/systemc_test$ cd build
saisri@DESKTOP-P73175QB:~/systemc_test/build$ cmake ..
-- Configuring done (0.1s)
-- Generating done (0.0s)
-- Build files have been written to: /home/saisri/systemc_test/build
saisri@DESKTOP-P73175QB:~/systemc_test/build$ make
[ 20%] Built target and_gate
[ 40%] Built target producer_consumer
[ 60%] Built target test_systemc
[ 80%] Built target tlm
[ 90%] Building CXX object CMakeFiles/binary_semaphore.dir/binary_semaphore.cpp.o
[100%] Linking CXX executable binary_semaphore
[100%] Built target binary_semaphore
saisri@DESKTOP-P73175QB:~/systemc_test/build$ ./binary_semaphore

        SystemC 2.3.4-Accellera --- Aug 21 2026 05:34:04
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        ALL RIGHTS RESERVED
Task1 Entered Critical Section at 1 s
Task1 Entered Critical Section at 4 s
Task1 Entered Critical Section at 7 s
saisri@DESKTOP-P73175QB:~/systemc_test/build$
```

```
[ 33%] Built target producer_consumer
[ 50%] Built target test_systemc
[ 66%] Built target tlm
[ 83%] Built target binary_semaphore
[ 91%] Building CXX object CMakeFiles/counting_semaphore.dir/counting_semaphore.cpp.o
[100%] Linking CXX executable counting_semaphore
[100%] Built target counting_semaphore
saisri@DESKTOP-P73175QB:~/systemc_test/buid$ ./counting_semaphore
-bash: ./counting_semaphore: No such file or directory
saisri@DESKTOP-P73175QB:~/systemc_test/build$ ./counting_semaphore

        SystemC 2.3.4-Accellera --- Aug 21 2026 05:34:04
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Warning: (W505) object already exists: Counting.task. Latter declaration will be renamed to Counting.task_0
In file: /home/saisri/systemc-2.3.4/src/sysc/kernel/sc_object_manager.cpp:153

Warning: (W505) object already exists: Counting.task. Latter declaration will be renamed to Counting.task_1
In file: /home/saisri/systemc-2.3.4/src/sysc/kernel/sc_object_manager.cpp:153
Task Accessing Resource at 1 s Remaining: 1
Task Accessing Resource at 1 s Remaining: 0
Task Accessing Resource at 3 s Remaining: 0
Task Accessing Resource at 4 s Remaining: 0
Task Accessing Resource at 5 s Remaining: 0
Task Accessing Resource at 6 s Remaining: 0
Task Accessing Resource at 7 s Remaining: 0
Task Accessing Resource at 8 s Remaining: 0
Task Accessing Resource at 9 s Remaining: 0
saisri@DESKTOP-P73175QB:~/systemc_test/build$
```

EXP 29: Priority-Based Interrupt Handling

```
Task Accessing Resource at 6 s Remaining: 0
Task Accessing Resource at 7 s Remaining: 0
Task Accessing Resource at 8 s Remaining: 0
Task Accessing Resource at 9 s Remaining: 0
saisri@DESKTOP-P73175QB:~/systemc_test/build$ cd ..
saisri@DESKTOP-P73175QB:~/systemc_test$ nano priority_interrupt.cpp
saisri@DESKTOP-P73175QB:~/systemc_test$ nano CMakeLists.txt
saisri@DESKTOP-P73175QB:~/systemc_test$ cd build
saisri@DESKTOP-P73175QB:~/systemc_test/build$ cmake ..
-- Configuring done (0.0s)
-- Generating done (0.0s)
-- Build files have been written to: /home/saisri/systemc_test/build
saisri@DESKTOP-P73175QB:~/systemc_test/build$ make
[ 14%] Built target and_gate
[ 28%] Built target producer_consumer
[ 42%] Built target test_systemc
[ 57%] Built target tlm
[ 71%] Built target binary_semaphore
[ 85%] Built target counting_semaphore
[ 92%] Building CXX object CMakeFiles/priority_interrupt.dir/priority_interrupt.cpp.o
[100%] Linking CXX executable priority_interrupt
[100%] Built target priority_interrupt
saisri@DESKTOP-P73175QB:~/systemc_test/build$ ./priority_interrupt

        SystemC 2.3.4-Accellera --- Aug 21 2026 05:34:04
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Low Priority Interrupt at 1 s
High Priority Interrupt at 2 s
saisri@DESKTOP-P73175QB:~/systemc_test/build$
```

```
-- Build files have been written to: /home/saisri/systemc_test/build
saisri@DESKTOP-P73175QB:~/systemc_test/build$ make
[ 12%] Built target and_gate
[ 25%] Built target producer_consumer
[ 37%] Built target test_systemc
[ 50%] Built target tlm
[ 62%] Built target binary_semaphore
[ 75%] Built target counting_semaphore
[ 87%] Built target priority_interrupt
[ 93%] Building CXX object CMakeFiles/rtos_scheduler.dir/rtos_scheduler.cpp.o
[100%] Linking CXX executable rtos_scheduler
[100%] Built target rtos_scheduler
saisri@DESKTOP-P73175QB:~/systemc_test/build$ ./rtos_scheduler

        SystemC 2.3.4-Accellera --- Aug 21 2026 05:34:04
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Task1 Running at 0 s
Task2 Running at 0 s
Task1 Running at 1 s
Task2 Running at 1 s
Task1 Running at 2 s
Task2 Running at 2 s
Task1 Running at 3 s
Task2 Running at 3 s
Task1 Running at 4 s
Task2 Running at 4 s
Task1 Running at 5 s
Task2 Running at 5 s
saisri@DESKTOP-P73175QB:~/systemc_test/build$
```